1. Evaluate the following integrals by integration by parts.

1). $\int xe^{-x}\,dx$

$u = x \quad dv = e^{-x}\,dx$

$du = dx \quad v = -e^{-x}$

$= -x \cdot e^{-x} - \int (-e^{-x})\,dx$

$= -x \cdot e^{-x} + \int e^{-x}\,dx$

$= -xe^{-x} - e^{-x} + C$

2). $\int t\sin(2t)\,dt$

$u = t \quad dv = \sin(2t)\,dt$

$du = dt \quad v = -\frac{1}{2}\cos(2t)$

$= -\frac{1}{2}t\cos(2t) - \int (-\frac{1}{2}\cos(2t))\,dt$

$= -\frac{1}{2}t\cos(2t) + \frac{1}{2}\int \cos(2t)\,dt$

$= -\frac{1}{2}t\cos(2t) + \frac{1}{4}\sin(2t) + C$

$u = \ln(2x+1) \quad dv = dx$

$du = \frac{2}{2x+1}$ [redacted]

3). $\int \ln(2x+1)\,dx$

$= x\ln(2x+1) - \int \frac{2x}{2x+1}\,dx$

$= x\ln(2x+1) - \int \left(\frac{2x+1}{2x+1} - \frac{1}{2x+1}\right)dx$

$= x\ln(2x+1) - \int \left(1 - \frac{1}{2x+1}\right)dx$

$= x\ln(2x+1) - x + \frac{1}{2}\ln|2x+1| + C$

4). $\int \sin^{-1}x\,dx$

$u = \sin^{-1}x \quad dv = dx$

$du = \frac{1}{\sqrt{1-x^2}}\,dx \quad v = x$

$= x\sin^{-1}x - \int \frac{x}{\sqrt{1-x^2}}\,dx$

see below

$= x\sin^{-1}x + \sqrt{1-x^2} + C$

$w = 1-x^2$

$dw = -2x\,dx$

$-\frac{1}{2}dw = x\,dx$

$\int \frac{x}{\sqrt{1-x^2}}\,dx = -\frac{1}{2}\int \frac{1}{\sqrt{w}}\,dw = -\frac{1}{2}\cdot 2\sqrt{w} + C$

$= -\sqrt{1-x^2} + C$

$u = t \qquad dv = \cos(3t)\,dt$

$du = dt \qquad v = \frac{1}{3}\sin(3t)$

5). $\displaystyle\int_0^{\frac{\pi}{6}} t\cos(3t)\,dt$

$= \frac{1}{3}t\sin(3t)\Big|_0^{\frac{\pi}{6}} - \int_0^{\pi/6} \frac{1}{3}\sin(3t)\,dt$

$= \left[\frac{1}{3}\cdot\frac{\pi}{6}\sin\left(\frac{\pi}{2}\right) - 0\right] + \frac{1}{9}\cos(3t)\Big|_0^{\frac{\pi}{6}}$

$= \frac{\pi}{18} + \frac{1}{9}\cos(2\pi) - \frac{1}{9}\cos(0)$

$= \frac{\pi}{18} - \frac{1}{9}$

6). $\displaystyle\int_1^2 \frac{\ln x}{x^2}\,dx.$

$u = \ln x \qquad du = \frac{1}{x^2}\,dx$

$du = \frac{1}{x}dx \qquad v = -\frac{1}{x}$

$= -\frac{1}{x}\ln x\Big|_1^2 - \int_1^2 \frac{1}{x}\cdot\left(-\frac{1}{x}\right)dx$

$= \left(-\frac{1}{2}\ln 2 + \ln 1\right) + \int_1^2 \frac{1}{x^2}\,dx$

$= -\frac{1}{2}\ln 2 - \frac{1}{x}\Big|_1^2$

$= -\frac{1}{2}\ln 2 - \left(\frac{1}{2} - 1\right)$

$= -\frac{1}{2}\ln 2 - \frac{1}{2} + 1$

$= \frac{1}{2} - \frac{1}{2}\ln 2$

$u = \ln x \qquad dv = x^4\,dx$

$du = \frac{1}{x}dx \qquad v = \frac{1}{5}x^5$

7). $\displaystyle\int x^4(\ln x)\,dx$

$= \frac{1}{5}x^5\ln x - \int \frac{1}{x}\cdot\frac{1}{5}x^5\,dx$

$= \frac{1}{5}x^5\ln x - \frac{1}{5}\int x^4\,dx$

$= \frac{1}{5}x^5\ln x - \frac{1}{5}\cdot\frac{1}{5}x^5 + C$

$= \frac{1}{5}x^5\ln x - \frac{1}{25}x^5 + C$

$u = \sin(3\theta) \qquad dv = e^{2\theta}\,d\theta$

$du = 3\cos(3\theta)\,d\theta \qquad v = \frac{1}{2}e^{2\theta}$

8). $\displaystyle\int e^{2\theta}\sin(3\theta)\,d\theta.$ (Denote by "I")

$= \frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{2}\int e^{2\theta}\cos 3\theta\,d\theta$

$u = \cos 3\theta \qquad dv = e^{2\theta}\,d\theta$

$du = -3\sin 3\theta \qquad v = \frac{1}{2}e^{2\theta}$

$= \frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{2}\left[\frac{1}{2}e^{2\theta}\cos 3\theta - \int \frac{1}{2}e^{2\theta}\cdot(-3\sin 3\theta)\,d\theta\right]$

$= \frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{4}e^{2\theta}\cos 3\theta - \frac{9}{4}\int e^{2\theta}\sin 3\theta\,d\theta$

$\Rightarrow \left(1 + \frac{9}{4}\right)\int e^{2\theta}\sin 3\theta\,d\theta = \frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{4}e^{2\theta}\cos 3\theta$

$\int e^{2\theta}\sin 3\theta\,d\theta = \frac{4}{13}\left(\frac{1}{2}e^{2\theta}\sin 3\theta - \frac{3}{4}e^{2\theta}\cos 3\theta\right) + C$

$= \frac{2}{13}e^{2\theta}\sin 3\theta - \frac{3}{13}e^{2\theta}\cos(3\theta) + C$